tec-speed 7.0 H-PK VT-770

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Precautions in Handling

Storage Condition & Shelf Life

		Prepreg		Laminate
Storage Condition	Temperature	Below 23°C (73°F)	Below 5°C (41°F)	Room
	Relative Humidity	Below 55%RH	/	/
Shelf Life		3 Months	6 Months	24 Months(airproof)

- The pre-preg exceeding shelf time should be retested.
- Take care in handling thin core laminates as they are easily damaged.
- If the pre-preg is not consumed within 48hrs after opening the vacuum package, it is recommended that the bags be resealed.
- Material is available in both long and short grain. The grain direction is indicated on the label with an arrow.

Designing and Inner layer Process

- Please be careful when single ply of 1078 or 1035, 1027 prepreg is designed to the dielectric layer.
- Baking of material is recommended for stabilization of dimension.
- In case of thin copper(ex.3um) with carrier copper foil, the carrier foil should be removed off before baking. Baking condition: Temp 200°C, 2 hours.
- Brown Oxide can be used. In the case of using Black Oxide,please check whether peel strength is acceptable for the usage.
- Holding time between brown oxide and press process: best control within 6 hours.

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Prepreg Availability

- E-Glass styles: 2116, 2113, 1078, 1067, 1037, etc.
- DK values are for impedance design.

100% PPT mil	Glass Style	R/C (%)	Dk					Df				
			@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 20GHz	@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 20GHz
1.7	1027	73	3.28	3.28	3.22	3.18	3.15	0.0030	0.0034	0.0037	0.0039	0.0041
1.8		75	3.25	3.24	3.20	3.16	3.13	0.0030	0.0034	0.0037	0.0039	0.0041
2.0		77	3.22	3.21	3.17	3.13	3.10	0.0029	0.0033	0.0036	0.0038	0.0040
1.9	1037	72	3.31	3.31	3.25	3.21	3.18	0.0030	0.0035	0.0038	0.0040	0.0042
2.0		74	3.27	3.27	3.23	3.19	3.16	0.0030	0.0034	0.0037	0.0039	0.0041
2.2		76	3.23	3.22	3.18	3.14	3.11	0.0029	0.0033	0.0036	0.0038	0.0040
2.5	1067	71	3.33	3.33	3.27	3.23	3.19	0.0030	0.0035	0.0038	0.0040	0.0042
2.6		73	3.28	3.28	3.22	3.18	3.15	0.0030	0.0034	0.0037	0.0039	0.0041
2.9		75	3.25	3.24	3.20	3.16	3.13	0.0030	0.0034	0.0037	0.0039	0.0041
3.1	1078	64	3.42	3.41	3.36	3.33	3.29	0.0031	0.0036	0.0039	0.0041	0.0043
3.3		66	3.40	3.40	3.35	3.31	3.27	0.0031	0.0036	0.0039	0.0041	0.0043
3.5		68	3.33	3.32	3.26	3.22	3.19	0.0030	0.0035	0.0038	0.0040	0.0042
4.0	2113	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045
5.0	2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045

Remarks:

- Press thickness test condition---Prepreg lamination size 18”*24”, Copper Foil---1oz/1oz, Flow---about 5%.
- Make laminated prepreg to micro-section and measure the thickness with microscope; This thickness is used for resistance design calculation.
- The thickness measured with micrometer is 0.2~0.4 mil larger than that measured with micro-section; and mainly used for total thickness design calculation.

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Laminates Availability

- E-Glass style.
- DK values are for impedance design.

Pressed Thickness		Glass Style	R/C (%)	Dk					Df				
µm	mil			@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 20GHz	@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 20GHz
43	1.7	1-1027	73	3.28	3.28	3.22	3.18	3.15	0.0030	0.0034	0.0037	0.0039	0.0041
50	2.0	1-1067	65	3.40	3.40	3.35	3.31	3.27	0.0031	0.0036	0.0039	0.0041	0.0043
64	2.5	1-1067	71	3.33	3.32	3.26	3.22	3.19	0.0030	0.0035	0.0038	0.0040	0.0042
76	3.0	1-1078	64	3.42	3.41	3.36	3.33	3.29	0.0031	0.0036	0.0039	0.0041	0.0043
89	3.5	1-1078	68	3.35	3.35	3.29	3.25	3.21	0.0030	0.0035	0.0038	0.0040	0.0042
102	4.0	1-3313	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045
102		2-1067	65	3.40	3.40	3.35	3.31	3.27	0.0031	0.0036	0.0039	0.0041	0.0043
127	5.0	1-2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045
127		2-1067	71	3.33	3.32	3.26	3.22	3.19	0.0030	0.0035	0.0038	0.0040	0.0042
152	6.0	2-1078	63	3.44	3.44	3.39	3.35	3.31	0.0031	0.0036	0.0039	0.0041	0.0043
203	8.0	2-3313	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045
254	10.0	2-2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045
305	12.0	3-3313	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045
356	14.0	2-2116+ 1-3313	55	3.49	3.49	3.44	3.40	3.36	0.0033	0.0038	0.0041	0.0043	0.0045
381	15.0	3-2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045
406	16.0	4-3313	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045
457	18.0	2-2116+ 2-3313	55	3.49	3.49	3.44	3.40	3.36	0.0033	0.0038	0.0041	0.0043	0.0045
508	20.0	4-2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045
635	25.0	5-2116	54	3.50	3.49	3.45	3.41	3.38	0.0033	0.0038	0.0041	0.0043	0.0045
711	28.0	7-3313	56	3.48	3.48	3.43	3.39	3.34	0.0033	0.0038	0.0041	0.0043	0.0045

- More types could be available upon request.

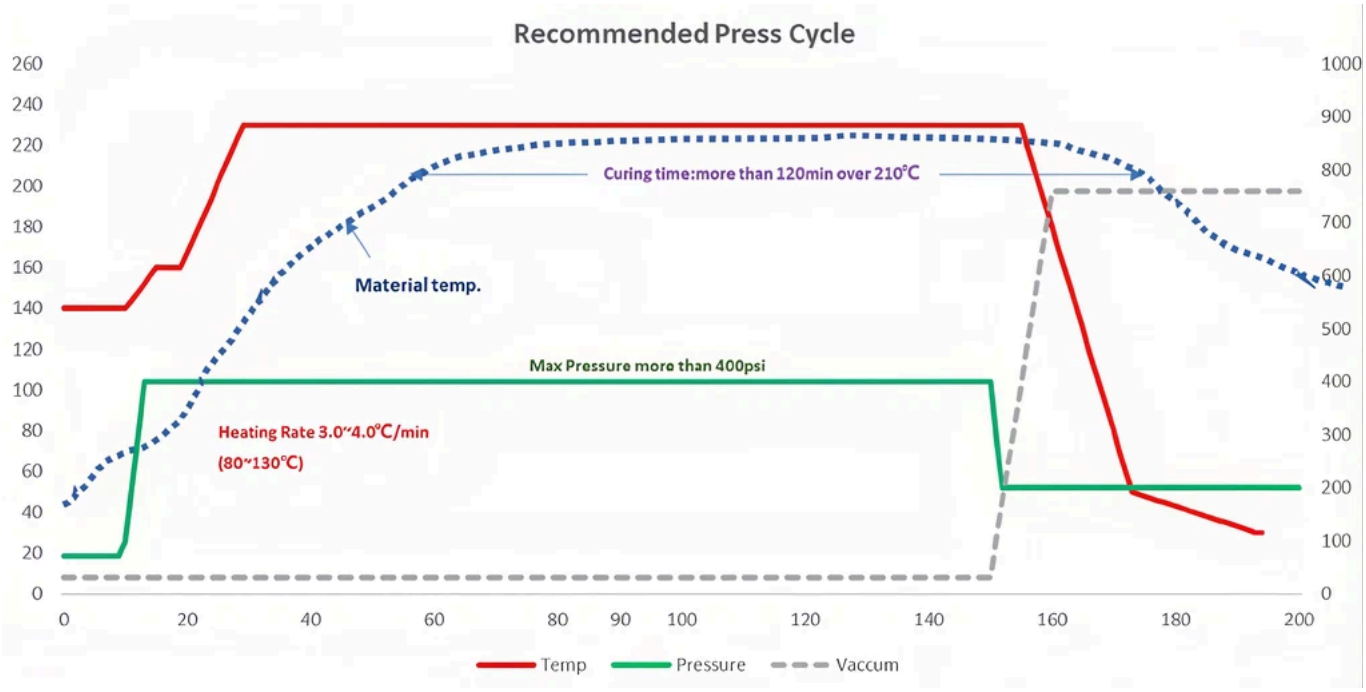
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Press Condition

1.Heating rate (Rate of Rise) of material [Material Temperature]: Programmable Press: ≥3°C/min.
2.Curing Temperature & Time: >120min at more than 210°C and peak temperature>215 °C.
3.Full Pressure: ≥400psi (28Kg/cm2) should be applied full pressure before 100 °C.
4.Vacuuming should be continued until over 140°C [Material Temperature].
5.Cushion for pressure evenness is needed.



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Typical Drilling Parameters

- Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, stack count and stack height etc.
- Please adjust drilling parameters after checking qualities of through holes.

Suggest Drilling parameter as blow:

Diameter (mm)	Spindle speed (KRPM)	Feed rate (IPM)	Retract rate (IPM)	Hit counts
0.10	200-300	25-38	800	1000
0.20	150-200	38-50	800	1000
0.30	100-150	38-60	800	1000
0.40	80-100	40-50	800	1000

- Smear,resin cracking or nail heading may occur if drilling condition doesn't match material.

Desmear Process

- The weight loss of VT-770 is less than as our conventional FR-4 material.
- We recommend a plasma cycle and a vertical permanganate cycle together to adequately prepare the hole walls for plating (If used by horizontal desmear, it should be need twice cycles) .

Plasma Desmear Parameter:
One time of FR-4 conditions is recommendable.

Permanganate Desmear Parameter:
Twice of FR-4 horizontal desmear condition is recommendable, or one time of FR-4 vertical desmear condition is recommendable.

Typical Permanganate Desmear parameter by Atotech chemical as below:

Process	Reagent type	Temperature(°C)	Duration(min)
Sewlling	Alkaline	60-70	5-10
Etching	Permanganate	70-80	10-15

Please consult the chemical supplier for suggested conditions.

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Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging, It is best to bake the PCBs at 125°C/4~6h before assembly before use.